

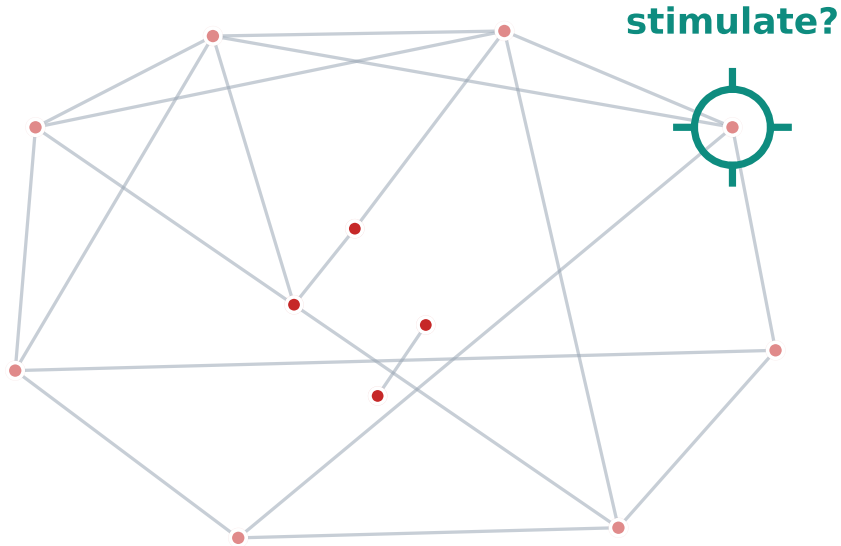
PERSONALISED MODELS OF RESTING-STATE fMRI · ALZHEIMER'S DISEASE

Optimal stimulation sites are not the most affected

C. Capone E. Cece A. Ciardiello G. Gigante E. Cisbani M. Mattia

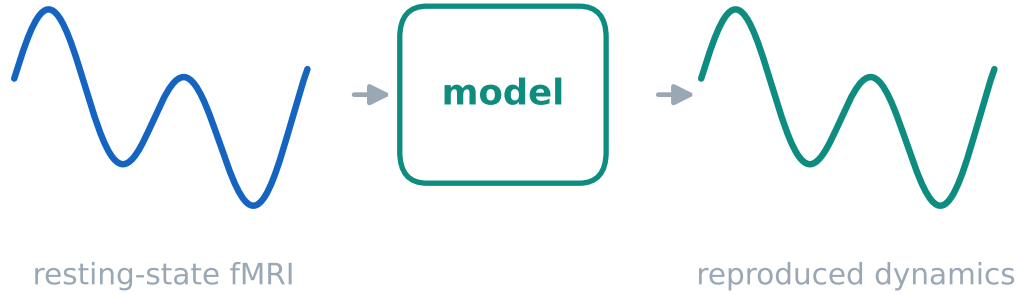
Istituto Superiore di Sanità, Rome

Where should we stimulate?



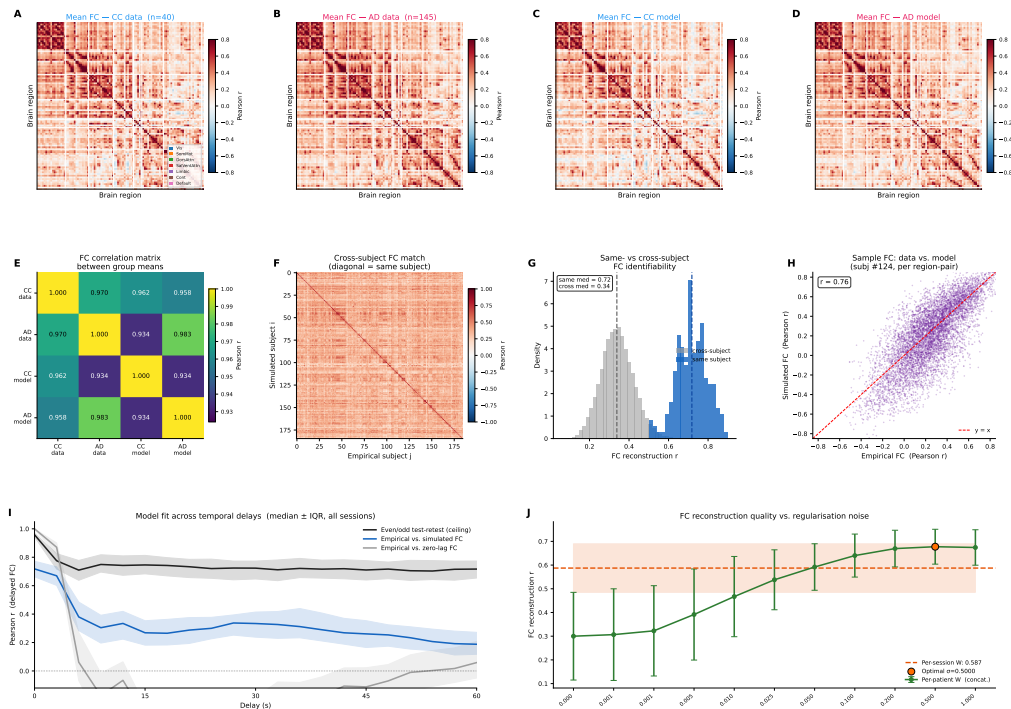
- Alzheimer's disrupts connectivity **brain-wide**
- Neuromodulation works best at a **single site**
- Is the signature reducible to a **focal target**?

A perturbable digital twin



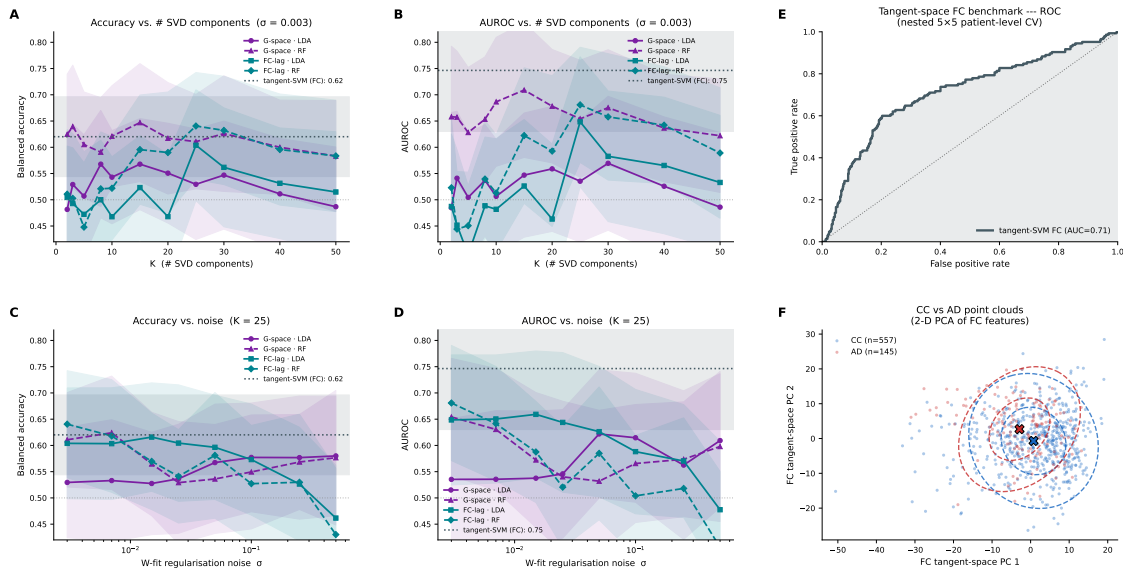
- One personalised model **per patient**
- Its free-running dynamics **reproduce** the patient's own
- We can now intervene **in silico**

The model is faithful and individual



- Reconstructs each subject's connectivity
- Identifiable **across** subjects
- A generative surrogate we can **modify**

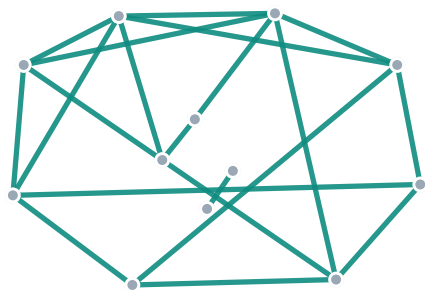
Reading the disease from the model



- Model parameters separate **controls** from **AD**
- Modest accuracy — below structural atrophy
- A **target** for stimulation, not a diagnostic

Two ways to intervene

perturb the graph

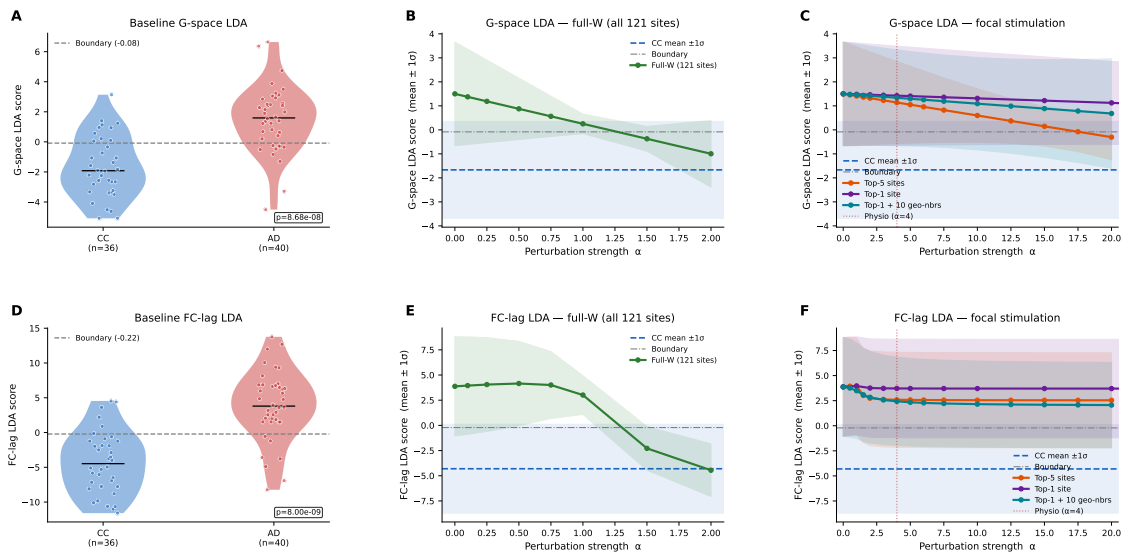


perturb the activity



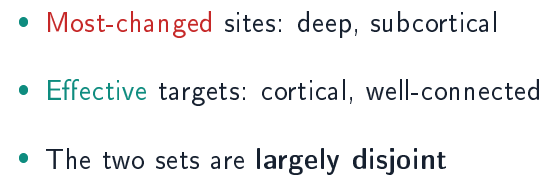
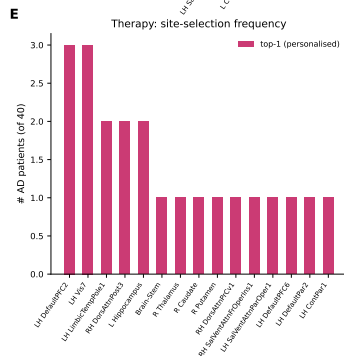
- **Graph:** reshape the connectivity itself
- **Activity:** a focal drive at one node
- They need not point to the **same target**

Correcting connectivity reverts the signature

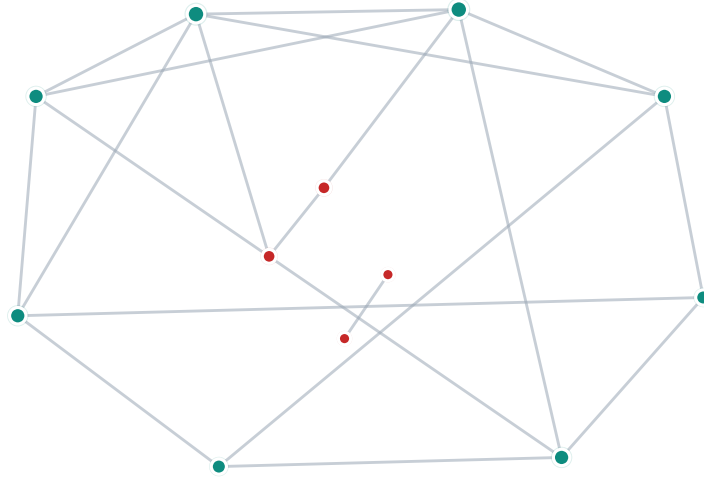


- Shifting the **whole** graph $\rightarrow$ reclassified as control
- Focal** edits fail at any amplitude
- The correction is **distributed**

A



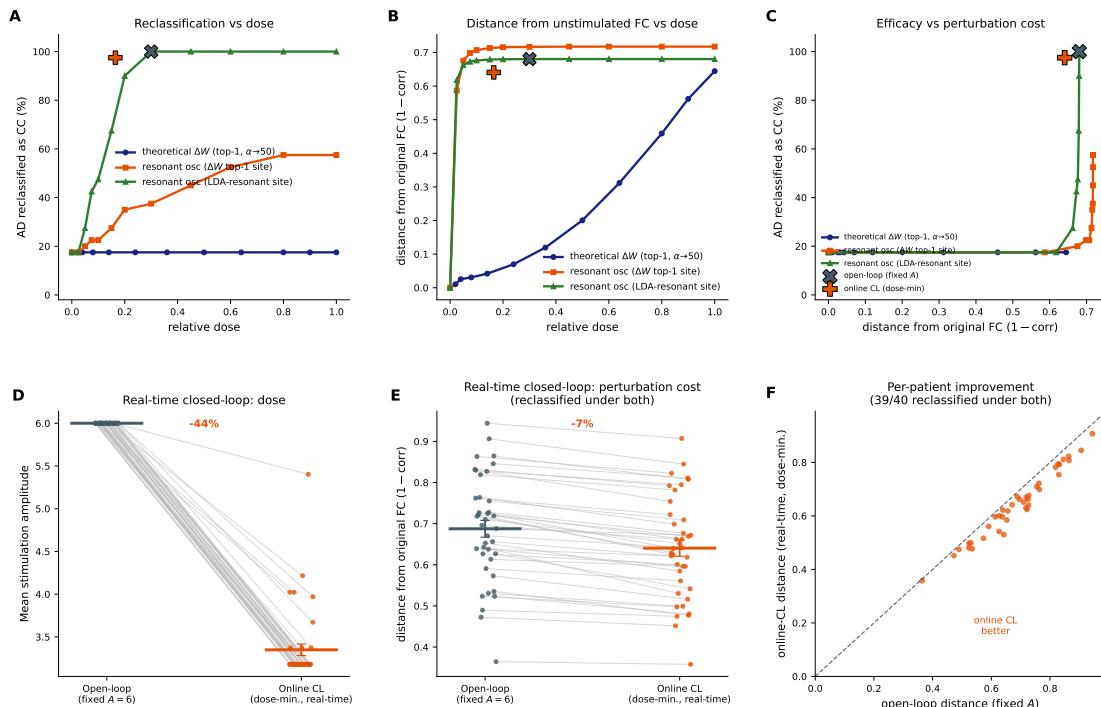
It follows from the connectivity



● well-connected hub ● weak node

- Most-affected nodes are weakly connected
- Weak nodes are **poor actuators**
- Hubs spread the correction network-wide

One site, and closed-loop control



- Resonant drive at the **right site** cures the patient
- A single read-out edit is **inert**
- Closed-loop titration $\rightarrow$ **lower dose**

Stimulate where the network **responds** —
not where it is most **affected**.

Model-informed, personalised targeting turns “where to stimulate” into a computable prescription.